

JEE MAIN REVISION MASTER: SOLUTIONS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 188 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Raoult's Law & Vapor Pressure (Ideal & Non-ideal)	124 PYQs	Vapor Pressure
Very High	Colligative Properties & Van 't Hoff Factor	66 PYQs	Colligative

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Raoult's Law & Vapor Pressure (Ideal & Non-ideal)	$P_{\text{total}} = P_A^{\text{circ}} x_A + P_B^{\text{circ}} x_B, \quad y_A = \frac{P_A}{P_{\text{total}}} = \frac{P_A^{\text{circ}} x_A}{P_{\text{total}}}, \quad P = K_H \cdot x \text{ (Henry)}$
Colligative Properties & Van 't Hoff Factor	$\frac{P^{\text{circ}} - P}{P^{\text{circ}}} = i \cdot x_B, \quad \Delta T_b = i K_b m, \quad \Delta T_f = i K_f m, \quad \Pi = i C R T, \quad i = 1 + (n - 1)\alpha$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Raoult's Law & Vapor Pressure (Ideal & Non-ideal) (Very High Yield — 124 PYQs)

Core Principle: Partial pressures of volatile mixtures, Dalton's law in vapor phase (y_A), and thermodynamic criteria for deviations ($\Delta H_{\text{(mix)}}$, $\Delta V_{\text{(mix)}}$).

■ **Caution / JEE Pitfall:** Mixture of Chloroform and Acetone exhibits **NEGATIVE** deviation from Raoult's law ($\Delta H_{\text{(mix)}} < 0$, $\Delta V_{\text{(mix)}} < 0$) due to intermolecular hydrogen bonding.

■ **Speed Trick:** Higher Henry's law constant K_H at a given pressure signifies **LOWER** solubility of the gas in liquid.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Given below are two statements : Given : Molar mass of C,H,O,Cl are 12,1,16 and 35.5gmol⁻¹ , respectively Statement I : In 30%(w/w..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "At T(K), 2 moles of liquid A and 3 moles of liquid B are mixed. The vapour pressure of ideal solution formed is 320 mm Hg . At this stage, ..." [Ans: C]
- [2026 Morning Shift] Q#1 (MCQ): " W ' g of a non-volatile electrolyte solid solute of molar mass ' M ' gmol⁻¹ when dissolved in 100 mL water, decreases vapour pressure..." [Ans: D]

3.2 Colligative Properties & Van 't Hoff Factor (Very High Yield — 66 PYQs)

Core Principle: Physical properties depending solely on solute particle count, cryoscopic/ebullioscopic constants, and Van 't Hoff factor i .

■ **Caution / JEE Pitfall:** When determining relative boiling/freezing points of salt solutions, compare effective molality $i \times m$, NOT just nominal molality m !

■ **Speed Trick:** For dimerization/association of n molecules: $i = 1 - \left(1 - \frac{1}{n}\right)\alpha$ implies $i < 1$ (e.g. benzoic acid in benzene, $i \approx 0.5$).

Representative Source PYQs:

- **[2026 Evening Shift] Q#1 (MCQ):** "At 27°C , 0.1M , 1L $\text{K}_4[\text{Fe}(\text{CN})_6]$ aqueous solution and 0.1M , 1L FeCl_3 aqueous solution are placed in a container separated by a s..." **[Ans: N/A]**
- **[2026 Evening Shift] Q#1 (Numerical):** "The osmotic pressure of a living cell is 12 atm at 300 K . The strength of sodium chloride solution that is isotonic with the living cell at ..." **[Ans: 15]**
- **[2026 Evening Shift] Q#2 (MCQ):** "Which of the following statements are not correct? A. For water, magnitude of K_b is more than the magnitude of K_f . B. The elevati..." **[Ans: N/A]**

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 28th January Morning Shift (Q#1)

At $T(\text{K})$, 2 moles of liquid A and 3 moles of liquid B are mixed. The vapour pressure of ideal solution formed is 320 mm Hg . At this stage, one mole of A and one mole of B are added to the solution. The vapour pressure is now measured as 328.6 mm Hg . The vapour pressure (in mm Hg) of A and B are respectively:

- (A) 400, 300
- (B) 600, 400
- (C) 500, 200 **[Correct]**
- (D) 300, 200

Correct Answer: C

Solution Analysis: For an ideal solution, by Raoult's law: $P_{\text{total}} = x_{\text{A}}P_{\text{A}}^0 + x_{\text{B}}P_{\text{B}}^0$ where $P_{\text{A}}^0, P_{\text{B}}^0$ are vapour pressures of pure liquids A and B. Step 1: First mixture (2 mol A + 3 mol B) Total moles = $2+3=5$ $x_{\text{A}} = (2/5), x_{\text{B}} = (3/5)$ Given $P_{\text{total}} = 320\text{ mm Hg}$: $320 = (2/5)P_{\text{A}}^0 + (3/5)P_{\text{B}}^0$ Multiply by 5: $1600 = 2P_{\text{A}}^0 + 3P_{\text{B}}^0$...(1) Step 2: After adding 1 mol A and 1 mol B New moles: A = 3, B = 4, total = 7 $x_{\text{A}} = (3/7), x_{\text{B}} = (4/7)$ Given new vapour pressure = 328.6 mm Hg : $328.6 = \frac{3}{7}P_{\text{A}}^0 + \frac{4}{7}P_{\text{B}}^0$

Benchmark Problem 2: JEE Main 2026 (Online) 21st January Evening Shift (Q#1)

The osmotic pressure of a living cell is 12 atm at 300 K . The strength of sodium chloride solution that is isotonic with the living cell at this temperature is _____ g L^{-1} . (Nearest integer) Given : $R = 0.08\text{ L atm K}^{-1}\text{ mol}^{-1}$ Assume complete dissociation of NaCl (Given: Molar mass of Na and Cl are 23 and 35.5 g mol^{-1} respectively.)

Correct Answer: 15

Solution Analysis: Osmotic pressure (π) = 12 atm Temperature (T) = 300 K Gas constant (R) = $0.08\text{ L atm K}^{-1}\text{ mol}^{-1}$ Sodium chloride (NaCl) is an electrolyte that breaks into ions in an aqueous solution: $\text{NaCl} \rightarrow \text{Na}^+ + \text{Cl}^-$ Since it is given to assume complete dissociation, one unit of NaCl produces 2 ions. Therefore, the van 't Hoff factor, $i=2$. According to the formula for osmotic pressure: $\pi = iCRT$ Substitute the given values into the equation: $12 = 2 \times C \times 0.08 \times 300$ $12 = 2 \times 24 \times C$ $12 = 48 \times C$